

Proofs of the Fourier-optics envelope identities in aberration-corrected LEEM

Wilson Lau

24 August 2026

Abstract

We give complete analytic proofs of the appendix identities of Yu, Lau and Altman, *Ultramicroscopy* **200** (2019) 160–168, for Fourier-optics (FO) image formation in aberration-corrected low-energy electron microscopy (ac-LEEM). After recording the modelling approximations that factor the bilinear kernel, we prove: the Gaussian characteristic functions that produce the spatial envelope (16); the complex Gaussian integral that produces the chromatic envelope (20)–(19); the polar form of the chromatic prefactor; the Gaussian algebra of the FO/CTF ratios (27); and the Jacobi–Anger expansion of a sinusoidal phase object. All statements are identities under the hypotheses named in the text. The first-order tilt truncation, the dropping of mixed tilt–energy terms, and the frozen-aperture approximation are *not* identities and are labelled as such. These proofs match the Lean 4 formalization in this repository.

Contents

1	Introduction	2
2	Notation and standing conventions	2
3	Gaussian calculus	3
4	Aberration identities	5
5	Spatial envelope	5
6	Chromatic envelope	6
7	Polar form of the chromatic envelope	8
8	Envelope ratios and the CTF comparison	9
9	Jacobi–Anger expansion of a sinusoidal object	10
10	Assembly	11
11	Correspondence with the Lean formalization	11
12	Analytic pitfalls	12

1 Introduction

Image formation in LEEM is bilinear in the object spectrum: partially coherent illumination mixes spatial frequencies $(\mathbf{q}, \mathbf{q}')$ rather than multiplying a single contrast transfer function (CTF). Yu, Lau and Altman [1] write the ensemble-averaged kernel as an integral of a coherent kernel against a Gaussian source density and a Gaussian energy density, then evaluate those integrals in Appendix A1.

This note proves every closed form used in that appendix and in the subsequent FO/CTF comparison, with the spatial-frequency convention of Tromp–Schramm [5, 3]: the wave aberration χ is measured in *waves*, the transfer factor is $W = e^{2\pi i \chi}$, and λ is restored so that q is a true spatial frequency. Object phase is in radians and is independent of that 2π convention. These identities are the forward kernel required to invert a through-focal intensity series for the exit wave; the inverse against this kernel is treated in a companion note.

What is proved. Gaussian Fourier transforms; FWHM–variance conversion; the gradient of χ_S ; the exact splitting $\Delta\chi_C = b_1\varepsilon + b_2\varepsilon^2$; closed forms of E_S and $E_{C,\text{tot}}$; polar form of E_{CC} ; $\Gamma_S = e^{-2\kappa uv}$; axis and perfect-coherence identities; $R_{\text{CTF}} = R_{\text{FO}} \Gamma_C \Gamma_S$; Jacobi–Anger.

What is not proved. Experimental claims, numerical accuracy of Table 1 parameters, and the modelling truncations of [1, Eqs. (3)–(4)].

2 Notation and standing conventions

Vectors are bold; $q = |\mathbf{q}|$. In one dimension, q, q', k are *signed* scalars, so qq' is a product of signed reals. The Fourier kernel is the unitary 2π -in-the-exponent convention

$$e^{2\pi i \mathbf{q} \cdot \mathbf{r}}, \quad d^2q = dq_x dq_y.$$

The principal branch of the complex logarithm has argument in $(-\pi, \pi]$, and $z^{1/2} = \exp(\frac{1}{2}\text{Log } z)$ with $z^{1/2} > 0$ for $z > 0$ real. The same branch defines $z^{-1/2}$.

Definition 2.1 (Object wave). $\psi_0(\mathbf{r}) = \sigma(\mathbf{r}) e^{i\phi(\mathbf{r})}$, with ϕ in radians.

Definition 2.2 (Spatial frequency). $\mathbf{q} = \boldsymbol{\alpha}/\lambda$, where $\boldsymbol{\alpha}$ is the emission-angle vector from the virtual object and λ is the wavelength after acceleration to column energy E .

Definition 2.3 (Aperture). For a circular contrast aperture of cutoff $q_{\text{ap}} = \alpha_{\text{ap}}/\lambda$,

$$M(\mathbf{q}) = \mathbf{1}_{\{|\mathbf{q}| \leq q_{\text{ap}}\}}.$$

In particular M is real, so $M^*(\mathbf{q}') = M(\mathbf{q}')$.

Approximation 2.4 (Frozen aperture). For a small source, $M(\mathbf{q} + \mathbf{k}) \approx M(\mathbf{q})$. This is a modelling hypothesis, not an identity.

Definition 2.5 (Spherical and defocus aberration). In waves,

$$\chi_S(\mathbf{q}, \Delta z) = \frac{1}{4}C_3\lambda^3q^4 + \frac{1}{6}C_5\lambda^5q^6 + \frac{1}{2}\Delta z \lambda q^2, \quad (1)$$

and $W_S = e^{2\pi i \chi_S}$. The sign of Δz is the Tromp–Schramm convention: $\Delta z > 0$ adds positive optical path for off-axis rays. Non-corrected (nac) LEEM keeps C_3 and drops C_5 ; aberration-corrected (ac) LEEM sets $C_3 = 0$ and keeps C_5 . In one dimension replace q^{2n} by the even power of the signed scalar.

Definition 2.6 (Chromatic aberration).

$$\chi_C(\mathbf{q}, \varepsilon) = \frac{1}{2} C_C \lambda \left(\frac{\varepsilon}{E} \right) q^2 + \frac{1}{2} C_{CC} \lambda \left(\frac{\varepsilon}{E} \right)^2 q^2 + \frac{1}{4} C_{3C} \lambda^3 \left(\frac{\varepsilon}{E} \right) q^4, \quad (2)$$

and $W_C = e^{2\pi i \chi_C}$. Nac keeps C_C and drops C_{CC}, C_{3C} ; ac sets $C_C = 0$ and keeps C_{CC}, C_{3C} .

Definition 2.7 (Coherent kernel). Paper Eq. (2):

$$R_0(\mathbf{q}, \mathbf{q}', \Delta z, \varepsilon) = M(\mathbf{q}) M(\mathbf{q}') W_S(\mathbf{q}, \Delta z) W_S^*(\mathbf{q}', \Delta z) W_C(\mathbf{q}, \varepsilon) W_C^*(\mathbf{q}', \varepsilon). \quad (3)$$

Lemma 2.8 (Unitary transfer). *If χ is real then $W(\mathbf{q}) W^*(\mathbf{q}') = e^{2\pi i (\chi(\mathbf{q}) - \chi(\mathbf{q}'))}$. At $\varepsilon = 0$, $\chi_C(\cdot, 0) = 0$, so $W_C(\cdot, 0) = 1$.*

Proof. $W = e^{2\pi i \chi}$ lies on the unit circle, hence $W^{-1} = W^*$. The second claim is $\chi_C(\mathbf{q}, 0) = 0$ by (2). $\square$

Definition 2.9 (Ensemble kernel). Paper Eq. (1):

$$R(\mathbf{q}, \mathbf{q}', \Delta z) = \int_{\mathbb{R}^2} d^2 k \int_{\mathbb{R}} d\varepsilon s(\mathbf{k}) c(\varepsilon) R_0(\mathbf{q} + \mathbf{k}, \mathbf{q}' + \mathbf{k}, \Delta z, \varepsilon). \quad (4)$$

Approximation 2.10 (First-order tilt, no mixed $k\varepsilon$). Expand $\chi_S(\mathbf{q} + \mathbf{k})$ to first order in $\mathbf{k}$ and drop mixed tilt–energy terms in $\chi_C(\mathbf{q} + \mathbf{k}, \varepsilon)$. Together with [Approximation 2.4](#) this is paper Eq. (4):

$$R_0(\mathbf{q} + \mathbf{k}, \mathbf{q}' + \mathbf{k}, \Delta z, \varepsilon) \approx R_0(\mathbf{q}, \mathbf{q}', \Delta z, 0) e^{2\pi i \mathbf{a} \cdot \mathbf{k}} e^{2\pi i (b_1 \varepsilon + b_2 \varepsilon^2)}, \quad (5)$$

with $\mathbf{a}, b_1, b_2$ as in [Section 4](#). The $(\mathbf{k}, \varepsilon)$ integral then factors as $E_S E_{C, \text{tot}}$. The remainder of this note evaluates those two integrals exactly.

3 Gaussian calculus

Definition 3.1 (Normalised Gaussians). For any width $\sigma > 0$ write

$$g_\sigma(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right).$$

The energy and source densities of the paper are

$$c(\varepsilon) = g_{\sigma_E}(\varepsilon), \quad (6)$$

$$s_{1D}(k) = g_{\sigma_{\text{ill}}}(k), \quad s(\mathbf{k}) = g_{\sigma_{\text{ill}}}(k_x) g_{\sigma_{\text{ill}}}(k_y) = \frac{1}{2\pi \sigma_{\text{ill}}^2} \exp\left(-\frac{|\mathbf{k}|^2}{2\sigma_{\text{ill}}^2}\right). \quad (7)$$

Lemma 3.2 (FWHM of a centred Gaussian). *Let $f(x) = \exp(-x^2/(2\sigma^2))$ with $\sigma > 0$. If $f(\pm\Delta/2) = \frac{1}{2}f(0)$, then*

$$\Delta = 2\sqrt{2 \ln 2} \sigma, \quad \sigma^2 = \frac{\Delta^2}{8 \ln 2}. \quad (8)$$

Proof. $f(0) = 1$, so the half-maximum condition is $\exp(-(\Delta/2)^2/(2\sigma^2)) = 1/2$. Taking $-\log$ yields $(\Delta/2)^2/(2\sigma^2) = \ln 2$, hence $\Delta^2/(8\sigma^2) = \ln 2$. The positive root is (8). Squaring $\Delta = 2\sqrt{2 \ln 2} \sigma$ recovers the variance formula. $\square$

The paper identifies ΔE with the FWHM of c and $q_{\text{ill}} = \alpha_{\text{ill}}/\lambda$ with the isotropic FWHM of s . Write $F := 2\sqrt{2 \ln 2}$ for the FWHM factor.

Lemma 3.3 (Product structure). $s(\mathbf{k}) = s_{1D}(k_x) s_{1D}(k_y)$.

Proof. $|\mathbf{k}|^2 = k_x^2 + k_y^2$ splits the exponential. The prefactors multiply: $(1/(\sigma_{\text{ill}}\sqrt{2\pi}))^2 = 1/(2\pi\sigma_{\text{ill}}^2)$ because $\sqrt{2\pi}\sqrt{2\pi} = 2\pi$. $\square$

Lemma 3.4 (Complex Gaussian integral). *Let $B, C, D \in \mathbb{C}$ with $\text{Re}(B) < 0$. Then*

$$\int_{\mathbb{R}} \exp(Bx^2 + Cx + D) dx = \left(\frac{\pi}{-B}\right)^{1/2} \exp\left(D - \frac{C^2}{4B}\right), \quad (9)$$

using the principal square root.

Proof. Complete the square: $Bx^2 + Cx + D = B(x + C/(2B))^2 + D - C^2/(4B)$. The hypothesis $\text{Re}(B) < 0$ gives Gaussian decay in a neighbourhood of the real axis, so the contour may be shifted and the integral equals $\exp(D - C^2/(4B))$ times $\int \exp(Bw^2) dw$. The standard evaluation of the latter is $(\pi/(-B))^{1/2}$. Because $\text{Re}(-B) > 0$, the number $-B$ lies in the open right half-plane, so $\text{Arg}(-B) \in (-\pi/2, \pi/2)$ and the principal square root never meets its branch cut (the negative reals). When B is real and negative this recovers the positive root $\sqrt{\pi/|B|}$. $\square$

Theorem 3.5 (One-dimensional characteristic function). *For $\sigma > 0$ and $a \in \mathbb{R}$,*

$$\int_{\mathbb{R}} g_{\sigma}(k) e^{2\pi i a k} dk = \exp(-2\pi^2 \sigma^2 a^2). \quad (10)$$

In particular this holds for $s_{1D} = g_{\sigma_{\text{ill}}}$ and for $c = g_{\sigma_E}$.

Proof. The integrand is a constant times $\exp(Bk^2 + Ck)$ with $B = -1/(2\sigma^2)$ (real) and $C = 2\pi i a$. Then $\text{Re}(B) < 0$. Lemma 3.4 gives the prefactor

$$\frac{1}{\sigma\sqrt{2\pi}} \left(\frac{\pi}{1/(2\sigma^2)}\right)^{1/2} = \frac{1}{\sigma\sqrt{2\pi}} \sqrt{2\pi\sigma^2} = 1,$$

using the positive square root ($\sigma > 0$). The exponent is $-C^2/(4B)$. Now $C^2 = (2\pi i a)^2 = -4\pi^2 a^2$, so

$$-\frac{C^2}{4B} = \frac{4\pi^2 a^2}{4B} = \frac{\pi^2 a^2}{B} = \pi^2 a^2 \cdot (-2\sigma^2) = -2\pi^2 \sigma^2 a^2.$$

Equivalently, the characteristic function of $\mathcal{N}(0, \sigma^2)$ is $\mathbb{E}[e^{itk}] = e^{-\sigma^2 t^2/2}$ at $t = 2\pi a$. $\square$

Theorem 3.6 (Two-dimensional characteristic function). *For $\sigma > 0$ and $\mathbf{a} \in \mathbb{R}^2$, write $s_{\sigma}(\mathbf{k}) = g_{\sigma}(k_x)g_{\sigma}(k_y)$. Then*

$$\int_{\mathbb{R}^2} s_{\sigma}(\mathbf{k}) e^{2\pi i \mathbf{a} \cdot \mathbf{k}} d^2 k = \exp(-2\pi^2 \sigma^2 |\mathbf{a}|^2). \quad (11)$$

The illumination density s is the case $\sigma = \sigma_{\text{ill}}$.

Proof. The density factors as a product of one-dimensional Gaussians of the same width, and $\mathbf{a} \cdot \mathbf{k} = a_x k_x + a_y k_y$ factors the exponential. The double integral is a product of two copies of Theorem 3.5, and the exponents add. $\square$

Corollary 3.7 (FWHM packaging). *If $q_{\text{ill}} = F\sigma_{\text{ill}}$ and $\kappa := \pi^2 q_{\text{ill}}^2/(4 \ln 2)$, then $2\pi^2 \sigma_{\text{ill}}^2 = \kappa$. Likewise $2\pi^2 \sigma_E^2 = \pi^2 \Delta E^2/(4 \ln 2)$.*

Proof. Substitute $\sigma^2 = \Delta^2/(8 \ln 2)$ from Lemma 3.2 into $2\pi^2 \sigma^2$:

$$2\pi^2 \cdot \frac{\Delta^2}{8 \ln 2} = \frac{\pi^2 \Delta^2}{4 \ln 2}.$$

$\square$

4 Aberration identities

Theorem 4.1 (Gradient of χ_S). *Viewing (1) as a radial C^∞ function on $\mathbb{R}^2$,*

$$\nabla_{\mathbf{q}}\chi_S(\mathbf{q}, \Delta z) = (C_3\lambda^3q^2 + C_5\lambda^5q^4 + \Delta z\lambda)\mathbf{q}. \quad (12)$$

In one dimension, $\partial_q\chi_S = C_3\lambda^3q^3 + C_5\lambda^5q^5 + \Delta z\lambda q$.

Proof. Write $\rho = q^2 = |\mathbf{q}|^2$, so $\chi_S = \frac{1}{4}C_3\lambda^3\rho^2 + \frac{1}{6}C_5\lambda^5\rho^3 + \frac{1}{2}\Delta z\lambda\rho$. The Euclidean gradient is $\nabla\rho = 2\mathbf{q}$. Chain rule: $\nabla(\rho^2) = 4\rho\mathbf{q} = 4q^2\mathbf{q}$ and $\nabla(\rho^3) = 6\rho^2\mathbf{q} = 6q^4\mathbf{q}$. The numerical coefficients cancel: $\frac{1}{4} \cdot 4 = 1$ and $\frac{1}{6} \cdot 6 = 1$ and $\frac{1}{2} \cdot 2 = 1$. In one dimension, differentiate the even powers q^4, q^6, q^2 directly; odd powers keep the sign of q . $\square$

Definition 4.2 (Tilt vector).

$$\mathbf{a}(\mathbf{q}, \mathbf{q}', \Delta z) := \nabla\chi_S(\mathbf{q}, \Delta z) - \nabla\chi_S(\mathbf{q}', \Delta z). \quad (13)$$

In one dimension, $a = C_3\lambda^3(q^3 - q'^3) + C_5\lambda^5(q^5 - q'^5) + \Delta z\lambda(q - q')$. On the diagonal, $\mathbf{a}(\mathbf{q}, \mathbf{q}) = 0$. On the axis $\mathbf{q}' = \mathbf{0}$, $\nabla\chi_S(\mathbf{0}) = 0$, so $\mathbf{a}(\mathbf{q}, \mathbf{0}) = \nabla\chi_S(\mathbf{q})$.

Theorem 4.3 (First-order remainder). *For each fixed $\mathbf{q}, \Delta z$,*

$$\chi_S(\mathbf{q} + \mathbf{k}, \Delta z) - \chi_S(\mathbf{q}, \Delta z) - \mathbf{k} \cdot \nabla\chi_S(\mathbf{q}, \Delta z) = o(|\mathbf{k}|) \quad (\mathbf{k} \rightarrow \mathbf{0}).$$

Proof. This is the definition of differentiability. Since χ_S is a polynomial the remainder is in fact $O(|\mathbf{k}|^2)$. The little-o statement is what paper Eq. (3) uses; it does *not* justify dropping the remainder uniformly in $\mathbf{q}$, which remains [Approximation 2.10](#). $\square$

Theorem 4.4 (b_1/b_2 splitting). *If $E \neq 0$, then $\chi_C(\mathbf{q}, \varepsilon) - \chi_C(\mathbf{q}', \varepsilon) = b_1\varepsilon + b_2\varepsilon^2$ with*

$$\begin{aligned} b_1(\mathbf{q}, \mathbf{q}') &= \frac{C_C\lambda}{2E}(q^2 - q'^2) + \frac{C_{3C}\lambda^3}{4E}(q^4 - q'^4), \\ b_2(\mathbf{q}, \mathbf{q}') &= \frac{C_{CC}\lambda}{2E^2}(q^2 - q'^2). \end{aligned} \quad (14)$$

In two dimensions, q^2 means $|\mathbf{q}|^2$. Units: b_1 is waves/energy, b_2 is waves/energy².

Proof. Subtract (2) at $\mathbf{q}$ and $\mathbf{q}'$ and collect powers of ε . The C_C term contributes $\frac{1}{2}C_C\lambda E^{-1}(q^2 - q'^2)\varepsilon$; the C_{3C} term contributes $\frac{1}{4}C_{3C}\lambda^3 E^{-1}(q^4 - q'^4)\varepsilon$; the C_{CC} term contributes $\frac{1}{2}C_{CC}\lambda E^{-2}(q^2 - q'^2)\varepsilon^2$. No other powers appear. This is an identity, not an approximation. $\square$

Corollary 4.5 (Specialisations). *On the diagonal, $b_1(\mathbf{q}, \mathbf{q}) = b_2(\mathbf{q}, \mathbf{q}) = 0$. Antisymmetry: $b_j(\mathbf{q}, \mathbf{q}') = -b_j(\mathbf{q}', \mathbf{q})$. Under ac ($C_C = 0$), $b_1 = (C_{3C}\lambda^3/(4E))(q^4 - q'^4)$. Under nac ($C_{CC} = C_{3C} = 0$), $b_2 = 0$ and $b_1 = (C_C\lambda/(2E))(q^2 - q'^2)$.*

Proof. Immediate from (14). $\square$

5 Spatial envelope

Definition 5.1 (Spatial envelope integral). After [Approximation 2.10](#),

$$E_S(\mathbf{q}, \mathbf{q}', \Delta z) := \int_{\mathbb{R}^2} s(\mathbf{k}) e^{2\pi i \mathbf{a} \cdot \mathbf{k}} d^2k, \quad (15)$$

and likewise in one dimension with s_{1D} and scalar a .

Theorem 5.2 (Closed form in σ_{ill}). *If $\sigma_{\text{ill}} > 0$, then $E_S = \exp(-2\pi^2\sigma_{\text{ill}}^2|\mathbf{a}|^2)$. The envelope is real, even under $(\mathbf{q}, \mathbf{q}') \leftrightarrow (\mathbf{q}', \mathbf{q})$, and equals 1 on the diagonal.*

Proof. Apply [Theorem 3.6](#) (or [Theorem 3.5](#) in one dimension). Reality: the exponent is real. Symmetry: $\mathbf{a}(\mathbf{q}', \mathbf{q}) = -\mathbf{a}(\mathbf{q}, \mathbf{q}')$, and $|\mathbf{a}|^2$ is even. On the diagonal $\mathbf{a} = \mathbf{0}$. $\square$

Theorem 5.3 (Paper Eq. (5)). *If $q_{\text{ill}} > 0$ and $\kappa = \pi^2 q_{\text{ill}}^2 / (4 \ln 2)$, then*

$$E_S(\mathbf{q}, \mathbf{q}', \Delta z) = \exp\left(-\frac{\pi^2 q_{\text{ill}}^2}{4 \ln 2} |\mathbf{a}(\mathbf{q}, \mathbf{q}', \Delta z)|^2\right) = e^{-\kappa |\mathbf{a}|^2}. \quad (16)$$

Proof. Combine [Theorem 5.2](#) with [Corollary 3.7](#). $\square$

Corollary 5.4 (Expanded $|\mathbf{a}|^2$). *Let $u(\mathbf{q}) := C_3 \lambda^3 q^2 + C_5 \lambda^5 q^4 + \Delta z \lambda$, so $\nabla \chi_S = u(\mathbf{q}) \mathbf{q}$ and $\mathbf{a} = u(\mathbf{q}) \mathbf{q} - u(\mathbf{q}') \mathbf{q}'$. Then*

$$|\mathbf{a}|^2 = u(\mathbf{q})^2 q^2 + u(\mathbf{q}')^2 q'^2 - 2u(\mathbf{q})u(\mathbf{q}') (\mathbf{q} \cdot \mathbf{q}').$$

The inner product $\mathbf{q} \cdot \mathbf{q}'$ cannot be replaced by $q q'$.

Proof. Expand $|\mathbf{x} - \mathbf{y}|^2$. $\square$

Corollary 5.5 (One-dimensional polynomial form). *In one dimension (16) is*

$$E_S(q, q', \Delta z) = \exp\left(-\kappa \left[C_3 \lambda^3 (q^3 - q'^3) + C_5 \lambda^5 (q^5 - q'^5) + \Delta z \lambda (q - q')\right]^2\right).$$

One may factor $(q - q')^2$ via $q^3 - q'^3 = (q - q')(q^2 + qq' + q'^2)$. For ac, $C_3 = 0$.

Corollary 5.6 (CTF slice). *$E_S(\mathbf{q}, \mathbf{0}, \Delta z) = \exp(-\kappa |\nabla \chi_S(\mathbf{q})|^2)$, the separable CTF spatial envelope.*

Proof. [Definition 4.2](#): $\mathbf{a}(\mathbf{q}, \mathbf{0}) = \nabla \chi_S(\mathbf{q})$. $\square$

6 Chromatic envelope

Definition 6.1 (Chromatic envelope integral).

$$E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}') := \int_{\mathbb{R}} c(\varepsilon) \exp(2\pi i(b_1 \varepsilon + b_2 \varepsilon^2)) d\varepsilon. \quad (17)$$

Lemma 6.2 (Quadratic coefficient). *The integrand of (17) is a constant times $\exp(B\varepsilon^2 + C\varepsilon)$ with*

$$B = -\frac{1}{2\sigma_E^2} + 2\pi i b_2, \quad C = 2\pi i b_1. \quad (18)$$

If $\sigma_E > 0$ then $\text{Re}(B) = -1/(2\sigma_E^2) < 0$, independently of b_2 . Writing $z := 1 - 4\pi i b_2 \sigma_E^2$ (so $\text{Re}(z) = 1 > 0$), one has $B = -z/(2\sigma_E^2)$.

Proof. Add the Gaussian exponent $-\varepsilon^2/(2\sigma_E^2)$ to the phase $2\pi i(b_1 \varepsilon + b_2 \varepsilon^2)$. The imaginary part of B is $2\pi b_2$ and does not affect $\text{Re}(B)$. The identity $B = -z/(2\sigma_E^2)$ is algebra:

$$-\frac{z}{2\sigma_E^2} = -\frac{1 - 4\pi i b_2 \sigma_E^2}{2\sigma_E^2} = -\frac{1}{2\sigma_E^2} + 2\pi i b_2.$$

Thus [Lemma 3.4](#) applies for every real b_2 as soon as $\sigma_E > 0$. $\square$

Lemma 6.3 (Square-root identity). *If $A > 0$ is real and $D \in \mathbb{C} \setminus \{0\}$ has $\text{Arg}(D) \neq \pi$, then $(A/D)^{1/2} = A^{1/2} D^{-1/2}$ on principal branches. This applies to $D = z = 1 - 4\pi i b_2 \sigma_E^2$, because $\text{Re}(z) = 1$ implies $\text{Arg}(z) \in (-\pi/2, \pi/2)$.*

Proof. $\text{Log}(A/D) = \log A - \text{Log } D$ provided $\text{Arg}(D) \neq \pi$ (the identity $\text{Log}(1/w) = -\text{Log } w$ fails on the negative reals). Exponentiate $\frac{1}{2}\text{Log}(A/D)$. $\square$

Theorem 6.4 (Paper Eq. (6b)). *Define*

$$E_{CC}(\mathbf{q}, \mathbf{q}') := (1 - 4\pi i b_2 \sigma_E^2)^{-1/2} = z^{-1/2} \quad (19)$$

on the principal branch, with value 1 at $b_2 = 0$. Then the Gaussian-integral prefactor of (17) equals E_{CC} , and $E_{CC}^2 = z^{-1}$.

Proof. Lemma 3.4 produces the prefactor

$$\frac{1}{\sigma_E \sqrt{2\pi}} \left(\frac{\pi}{-B} \right)^{1/2}.$$

From Lemma 6.2, $\pi/(-B) = \pi \cdot (2\sigma_E^2)/z = 2\pi\sigma_E^2/z$. Lemma 6.3 with $A = 2\pi\sigma_E^2 > 0$ gives $(\pi/(-B))^{1/2} = (2\pi\sigma_E^2)^{1/2} z^{-1/2}$. The positive real root $(2\pi\sigma_E^2)^{1/2} = \sigma_E \sqrt{2\pi}$ cancels the Gaussian normalisation, leaving $z^{-1/2} = E_{CC}$. At $b_2 = 0$, $z = 1$ and $1^{-1/2} = 1$ (not -1). The identity $E_{CC}^2 = z^{-1}$ is $z^{-1/2} z^{-1/2} = z^{-1}$, valid because $\text{Arg}(z) \neq \pi$. $\square$

In FWHM variables, $4\pi b_2 \sigma_E^2 = \pi b_2 \Delta E^2 / (2 \ln 2)$, so $z = 1 - i(\pi b_2 \Delta E^2) / (2 \ln 2)$.

Theorem 6.5 (Paper Eq. (6a)). *If $\sigma_E > 0$, then*

$$E_{C,\text{tot}} = E_{CC} \exp(-2\pi^2 \sigma_E^2 b_1^2 E_{CC}^2) = (1 - 4\pi i b_2 \sigma_E^2)^{-1/2} \exp\left(-\frac{2\pi^2 \sigma_E^2 b_1^2}{1 - 4\pi i b_2 \sigma_E^2}\right). \quad (20)$$

Proof. Lemma 3.4 supplies the exponent $-C^2/(4B)$ with $C = 2\pi i b_1$. Then $C^2 = (2\pi i b_1)^2 = -4\pi^2 b_1^2$, hence

$$-\frac{C^2}{4B} = \frac{4\pi^2 b_1^2}{4B} = \frac{\pi^2 b_1^2}{B}.$$

Substitute $B = -z/(2\sigma_E^2)$:

$$\frac{\pi^2 b_1^2}{B} = \pi^2 b_1^2 \cdot \frac{-2\sigma_E^2}{z} = -2\pi^2 \sigma_E^2 b_1^2 / z.$$

Since $E_{CC}^2 = z^{-1}$, this is $-2\pi^2 \sigma_E^2 b_1^2 E_{CC}^2$. Multiply by the prefactor of Theorem 6.4. $\square$

Remark 6.6 (Hanßen–Trepte packaging). One has $2\pi^2 \sigma_E^2 b_1^2 = (\Delta E)^2 (2\pi b_1)^2 / (16 \ln 2)$, where $2\pi b_1$ is the coefficient of ε in the *radian* phase $2\pi \Delta \chi_C$. Both π (from $W = e^{2\pi i \chi}$) and $16 \ln 2$ (from FWHM) must appear.

Corollary 6.7 (Nac / $b_2 = 0$). *If $b_2 = 0$ then $E_{CC} = 1$ and $E_{C,\text{tot}} = \exp(-2\pi^2 \sigma_E^2 b_1^2)$. On the nac CTF slice $\mathbf{q}' = \mathbf{0}$, $b_1 = C_C \lambda q^2 / (2E)$, hence*

$$E_{C,\text{tot}}(\mathbf{q}, \mathbf{0}) = \exp\left(-\frac{(\pi C_C \lambda \Delta E q^2)^2}{16 \ln 2 E^2}\right), \quad (21)$$

the standard chromatic envelope of Schramm et al. [3].

Proof. Theorem 6.5 with $z = 1$, then substitute b_1 and Corollary 3.7:

$$2\pi^2 \sigma_E^2 b_1^2 = \frac{\pi^2 \Delta E^2}{4 \ln 2} \cdot \frac{C_C^2 \lambda^2 q^4}{4E^2} = \frac{(\pi C_C \lambda \Delta E q^2)^2}{16 \ln 2 E^2}.$$

$\square$

Corollary 6.8 (Conjugation). $E_{C,\text{tot}}(-b_1, -b_2) = \overline{E_{C,\text{tot}}(b_1, b_2)}$. Equivalently $E_{C,\text{tot}}(\mathbf{q}', \mathbf{q}) = \overline{E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}')}$.

Proof. The density c is real, and $\exp(2\pi i \psi)^* = \exp(-2\pi i \psi)$ for real ψ . Thus the integrand at $(-b_1, -b_2)$ is the conjugate of the integrand at (b_1, b_2) . Integrate, or apply Corollary 4.5 to swap $(\mathbf{q}, \mathbf{q}')$. $\square$

7 Polar form of the chromatic envelope

Write $y := 4\pi b_2 \sigma_E^2 = \pi b_2 \Delta E^2 / (2 \ln 2)$, so $E_{CC} = (1 - iy)^{-1/2}$. Under ac, $y = \gamma(q^2 - q'^2)$ with

$$\gamma := \frac{\pi C_{CC} \lambda \Delta E^2}{4 \ln 2 E^2}. \quad (22)$$

Lemma 7.1 (Modulus and argument of $1 - iy$). *For $y \in \mathbb{R}$, $|1 - iy| = \sqrt{1 + y^2}$ and $\text{Arg}(1 - iy) = -\arctan y$. In particular $\text{Arg}(1 - iy) \in (-\pi/2, \pi/2)$.*

Proof. Real part $1 > 0$, imaginary part $-y$. The modulus is $\sqrt{1 + y^2}$. For a number in the closed right half-plane, $\text{Arg} = \text{atan2}(\text{Im}, \text{Re}) = \text{atan2}(-y, 1) = -\arctan y$, where $\arctan : \mathbb{R} \rightarrow (-\pi/2, \pi/2)$. Because the real part is identically 1, the argument never jumps. $\square$

Theorem 7.2 (Polar form of E_{CC}). *On the principal branch,*

$$(1 - iy)^{-1/2} = (1 + y^2)^{-1/4} \exp\left(\frac{i}{2} \arctan y\right). \quad (23)$$

Hence under ac,

$$E_{CC} = [1 + \gamma^2(q^2 - q'^2)^2]^{-1/4} \exp\left(\frac{i}{2} \arctan(\gamma(q^2 - q'^2))\right).$$

Proof. $z^{-1/2} = |z|^{-1/2} \exp(-i \text{Arg}(z)/2)$. By Lemma 7.1, $|z|^{-1/2} = (1 + y^2)^{-1/4}$ and $-\text{Arg}(z)/2 = \frac{1}{2} \arctan y$. The phase of E_{CC} therefore lies in $(-\pi/4, \pi/4)$ and never jumps. At $y = 0$ one recovers $+1$. $\square$

Lemma 7.3 (Polar form of $1/z$). $1/(1 - iy) = (1 + iy)/(1 + y^2)$.

Proof. Multiply numerator and denominator by $1 + iy$. The denominator is $|1 - iy|^2 = 1 + y^2$. $\square$

Theorem 7.4 (Polar form of the remaining exponential, ac). *Let*

$$\eta := \frac{(\pi C_{3C} \lambda^3 \Delta E)^2}{64 \ln 2 E^2}, \quad (24)$$

so that $2\pi^2 \sigma_E^2 b_1^2 = \eta(q^4 - q'^4)^2$ under ac. Then

$$\begin{aligned} \exp(-2\pi^2 \sigma_E^2 b_1^2 E_{CC}^2) &= \exp\left(-\frac{\eta(q^4 - q'^4)^2}{1 + \gamma^2(q^2 - q'^2)^2}\right) \\ &\cdot \exp\left(-i \frac{\eta \gamma(q^2 - q'^2)(q^4 - q'^4)^2}{1 + \gamma^2(q^2 - q'^2)^2}\right). \end{aligned} \quad (25)$$

Proof. Under ac, Corollary 4.5 gives $b_1 = (C_{3C} \lambda^3 / (4E))(q^4 - q'^4)$, so

$$2\pi^2 \sigma_E^2 b_1^2 = \frac{\pi^2 \Delta E^2}{4 \ln 2} \cdot \frac{C_{3C}^2 \lambda^6}{16 E^2} (q^4 - q'^4)^2 = \eta(q^4 - q'^4)^2.$$

Now $E_{CC}^2 = 1/z = 1/(1 - iy)$. The exponent is $-2\pi^2 \sigma_E^2 b_1^2 \cdot (1 + iy)/(1 + y^2)$ by Lemma 7.3. The real part is $-\eta(q^4 - q'^4)^2/(1 + y^2)$; the imaginary part is $-\eta(q^4 - q'^4)^2 y/(1 + y^2)$. Substitute $y = \gamma(q^2 - q'^2)$. $\square$

Theorem 7.5 (Amplitude and phase of $E_{C,\text{tot}}$, ac). *Write $E_{C,\text{tot}} = A e^{i\theta}$ with $A \geq 0$. Then*

$$\begin{aligned} A(q, q') &= [1 + \gamma^2(q^2 - q'^2)^2]^{-1/4} \exp\left(-\frac{\eta(q^4 - q'^4)^2}{1 + \gamma^2(q^2 - q'^2)^2}\right), \\ \theta(q, q') &= \frac{1}{2} \arctan(\gamma(q^2 - q'^2)) - \frac{\eta \gamma(q^2 - q'^2)(q^4 - q'^4)^2}{1 + \gamma^2(q^2 - q'^2)^2}. \end{aligned} \quad (26)$$

Parity: $A(q, q') = A(q', q) = A(|q|, |q'|)$ and $\theta(q, q') = -\theta(q', q)$.

Proof. Product of Theorems 7.2 and 7.4: moduli multiply, arguments add. Parity follows because y and $(q^4 - q'^4)$ both change sign under $(q, q') \leftrightarrow (q', q)$, while y^2 and $(q^4 - q'^4)^2$ do not. $\square$

8 Envelope ratios and the CTF comparison

Definition 8.1 (Paper Eqs. (7a)–(7b)).

$$\begin{aligned}\Gamma_C(\mathbf{q}, \mathbf{q}') &= \frac{E_{C,\text{tot}}(\mathbf{q}, \mathbf{0}) E_{C,\text{tot}}(\mathbf{q}', \mathbf{0})}{E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}')}, \\ \Gamma_S(\mathbf{q}, \mathbf{q}', \Delta z) &= \frac{E_S(\mathbf{q}, \mathbf{0}, \Delta z) E_S(\mathbf{q}', \mathbf{0}, \Delta z)}{E_S(\mathbf{q}, \mathbf{q}', \Delta z)}.\end{aligned}\tag{27}$$

Printed (7a) uses the *non-conjugated* product $E_{C,\text{tot}}(\mathbf{q}, \mathbf{0}) E_{C,\text{tot}}(\mathbf{q}', \mathbf{0})$. For ac, $E_{C,\text{tot}}$ is complex, so Γ_C is complex; the plotted quantity is $|\Gamma_C|$.

Theorem 8.2 (Γ_S algebra). *Let $\mathbf{u} = \nabla \chi_S(\mathbf{q}, \Delta z)$, $\mathbf{v} = \nabla \chi_S(\mathbf{q}', \Delta z)$, and $\kappa = \pi^2 q_{\text{ill}}^2 / (4 \ln 2)$. Then*

$$\Gamma_S = \exp(-2\kappa \mathbf{u} \cdot \mathbf{v}).\tag{28}$$

In one dimension, $\Gamma_S = \exp(-2\kappa (\partial_q \chi_S(q))(\partial_q \chi_S(q')))$.

Proof. $E_S = e^{-\kappa |\mathbf{a}|^2}$ with $\mathbf{a} = \mathbf{u} - \mathbf{v}$, and $\nabla \chi_S(\mathbf{0}) = \mathbf{0}$, so $E_S(\mathbf{q}, \mathbf{0}) = e^{-\kappa |\mathbf{u}|^2}$ and $E_S(\mathbf{q}', \mathbf{0}) = e^{-\kappa |\mathbf{v}|^2}$. Therefore

$$\Gamma_S = \exp(-\kappa |\mathbf{u}|^2 - \kappa |\mathbf{v}|^2 + \kappa |\mathbf{u} - \mathbf{v}|^2).$$

Expand $|\mathbf{u} - \mathbf{v}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 - 2\mathbf{u} \cdot \mathbf{v}$. The quadratic terms cancel, leaving $e^{-2\kappa \mathbf{u} \cdot \mathbf{v}}$. $\square$

Corollary 8.3 (One-dimensional ac formula). *If $C_3 = 0$ and $q_{\text{ill}} > 0$, then*

$$\Gamma_S(q, q', \Delta z) = \exp\left[-\frac{\pi^2 q_{\text{ill}}^2}{2 \ln 2} (C_5 \lambda^5 q^5 + \Delta z \lambda q) (C_5 \lambda^5 q'^5 + \Delta z \lambda q')\right].\tag{29}$$

Proof. [Theorem 8.2](#) with $2\kappa = \pi^2 q_{\text{ill}}^2 / (2 \ln 2)$ and [Theorem 4.1](#) with $C_3 = 0$. $\square$

Theorem 8.4 (Perfect coherence). *If $q_{\text{ill}} = \Delta E = 0$ (equivalently $\sigma_{\text{ill}} = \sigma_E = 0$), then every envelope equals 1, hence $\Gamma_C = \Gamma_S = 1$ and $\Gamma_C \Gamma_S = 1$ for all $(\mathbf{q}, \mathbf{q}', \Delta z)$.*

Proof. $\exp(0) = 1$ and $1^{-1/2} = 1$. The closed forms extend by continuity to vanishing widths. $\square$

Theorem 8.5 (Axes). $\Gamma_S(\mathbf{q}, \mathbf{0}, \Delta z) = \Gamma_S(\mathbf{0}, \mathbf{q}', \Delta z) = 1$. $\Gamma_C(\mathbf{q}, \mathbf{0}) = 1$. *If $\Delta E > 0$ then $|\Gamma_C(\mathbf{0}, \mathbf{q}')| = 1$.*

Proof. Write $E_S(\mathbf{q}, \mathbf{q}') for $E_S(\mathbf{q}, \mathbf{q}', \Delta z)$. Then $E_S(\mathbf{0}, \mathbf{0}) = 1$ because $\mathbf{a}(\mathbf{0}, \mathbf{0}) = \mathbf{0}$. Hence$

$$\Gamma_S(\mathbf{q}, \mathbf{0}, \Delta z) = \frac{E_S(\mathbf{q}, \mathbf{0}) E_S(\mathbf{0}, \mathbf{0})}{E_S(\mathbf{q}, \mathbf{0})} = 1,$$

and likewise $\Gamma_S(\mathbf{0}, \mathbf{q}', \Delta z) = 1$ by symmetry of E_S .

For the chromatic ratios, $b_1(\mathbf{0}, \mathbf{0}) = b_2(\mathbf{0}, \mathbf{0}) = 0$, so $E_{C,\text{tot}}(\mathbf{0}, \mathbf{0}) = 1$. Therefore

$$\Gamma_C(\mathbf{q}, \mathbf{0}) = \frac{E_{C,\text{tot}}(\mathbf{q}, \mathbf{0}) E_{C,\text{tot}}(\mathbf{0}, \mathbf{0})}{E_{C,\text{tot}}(\mathbf{q}, \mathbf{0})} = 1.$$

For the other axis, $b_j(\mathbf{0}, \mathbf{q}') = -b_j(\mathbf{q}', \mathbf{0})$, so [Corollary 6.8](#) gives $E_{C,\text{tot}}(\mathbf{0}, \mathbf{q}') = \overline{E_{C,\text{tot}}(\mathbf{q}', \mathbf{0})}$. Let $Z := E_{C,\text{tot}}(\mathbf{q}', \mathbf{0})$. Then $Z \neq 0$ ($E_{CC} \neq 0$ and the exponential never vanishes), and

$$\Gamma_C(\mathbf{0}, \mathbf{q}') = \frac{E_{C,\text{tot}}(\mathbf{0}, \mathbf{0}) Z}{\overline{Z}} = \frac{Z}{\overline{Z}},$$

which has modulus 1. (For ac this need not equal 1: only the modulus is identically one.) $\square$

Definition 8.6 (Working FO and CTF kernels). After [Approximation 2.10](#),

$$R_{\text{FO}}(\mathbf{q}, \mathbf{q}', \Delta z) := R_0(\mathbf{q}, \mathbf{q}', \Delta z, 0) E_S(\mathbf{q}, \mathbf{q}', \Delta z) E_{C, \text{tot}}(\mathbf{q}, \mathbf{q}'), \quad (30)$$

$$R_{\text{CTF}}(\mathbf{q}, \mathbf{q}', \Delta z) := R_0(\mathbf{q}, \mathbf{q}', \Delta z, 0) E_S(\mathbf{q}, \mathbf{0}, \Delta z) E_S(\mathbf{q}', \mathbf{0}, \Delta z) E_{C, \text{tot}}(\mathbf{q}, \mathbf{0}) E_{C, \text{tot}}(\mathbf{q}', \mathbf{0}), \quad (31)$$

where the last two chromatic factors are *not* conjugated, matching printed (7a). Spatial envelopes are real, so conjugating $E_S(\mathbf{q}', \mathbf{0}, \Delta z)$ is redundant. The spatial CTF factors still depend on defocus.

Lemma 8.7 (Hermiticity of R_0). $\overline{R_0(\mathbf{q}, \mathbf{q}', \Delta z, \varepsilon)} = R_0(\mathbf{q}', \mathbf{q}, \Delta z, \varepsilon)$.

Proof. M is real. Each of W_S and W_C is unimodular, so $W^*(\mathbf{q})W(\mathbf{q}') = \overline{W(\mathbf{q})W^*(\mathbf{q}')}$ for $W \in \{W_S, W_C\}$. The product of these factors therefore swaps $(\mathbf{q}, \mathbf{q}')$ under conjugation. $\square$

Theorem 8.8 (FO recovers CTF on the axis). $R_{\text{FO}}(\mathbf{q}, \mathbf{0}, \Delta z) = R_{\text{CTF}}(\mathbf{q}, \mathbf{0}, \Delta z)$.

Proof. On $\mathbf{q}' = \mathbf{0}$, [Theorem 8.5](#) gives $\Gamma_C = \Gamma_S = 1$, or expand (30)–(31) using $E_S(\mathbf{0}, \mathbf{0}, \Delta z) = 1$ and $E_{C, \text{tot}}(\mathbf{0}, \mathbf{0}) = 1$. $\square$

Theorem 8.9 (Paper comparison identity).

$$\boxed{R_{\text{CTF}} = R_{\text{FO}} \Gamma_C \Gamma_S}. \quad (32)$$

Proof. Neither envelope vanishes: E_S is an exponential, and $E_{C, \text{tot}} = E_{CC}$ times an exponential with $E_{CC} = z^{-1/2}$ and $z \neq 0$ ($\text{Re}(z) = 1$). Multiplying (30) by (27) therefore cancels $E_S(\mathbf{q}, \mathbf{q}')$ and $E_{C, \text{tot}}(\mathbf{q}, \mathbf{q}')$, leaving (31). $\square$

Thus $\Gamma_C \Gamma_S \neq 1$ is precisely the warning that a separable CTF kernel differs from the FO kernel.

9 Jacobi–Anger expansion of a sinusoidal object

A kinematic LEEM phase object of physical amplitude A and wavelength Λ is $\psi_0(x) \propto \exp(i\varphi \sin(kx))$ with $\varphi = 4\pi A/\lambda_0$ and $k = 2\pi/\Lambda$. The phase amplitude is a definition of the reflection kinematics, not an FO theorem.

Definition 9.1 (Fourier–Bessel coefficients). For $\varphi \in \mathbb{R}$ and $n \in \mathbb{Z}$,

$$J_n(\varphi) := \frac{1}{2\pi} \int_{(0, 2\pi]} \exp(i\varphi \sin \theta) e^{-in\theta} d\theta. \quad (33)$$

Lemma 9.2 (Periodicity and unit modulus). $\theta \mapsto \exp(i\varphi \sin \theta)$ is 2π -periodic and has modulus 1. Its second derivative is bounded by $|\varphi| + \varphi^2$.

Proof. Sine is 2π -periodic. The exponent $i\varphi \sin \theta$ is purely imaginary, so the exponential has modulus 1. Differentiating, $f' = i\varphi \cos \theta \cdot f$ and $f'' = i\varphi(-\sin \theta)f + (i\varphi \cos \theta)^2 f$, hence $|f''| \leq |\varphi| + \varphi^2$. $\square$

Lemma 9.3 (Bessel bound). If $n \neq 0$ then $|J_n(\varphi)| \leq (|\varphi| + \varphi^2)/n^2$. Consequently $\sum_{n \in \mathbb{Z}} J_n(\varphi)$ is absolutely summable.

Proof. Two integrations by parts on the circle (periodic endpoints cancel) express J_n , for $n \neq 0$, as the Fourier coefficient of f'' divided by $(in)^2$. The bound on $|f''|$ yields the coefficient bound. Comparison with $\sum n^{-2}$ gives summability; the $n = 0$ term is finite. $\square$

Theorem 9.4 (Jacobi–Anger). *For all real φ, θ ,*

$$\exp(i\varphi \sin \theta) = \sum_{n \in \mathbb{Z}} J_n(\varphi) e^{in\theta}, \quad (34)$$

the series converging absolutely and uniformly on $\mathbb{R}$. Hence $\exp(i\varphi \sin(kx)) = \sum_n J_n(\varphi) e^{inkx}$.

Proof. The map $\theta \mapsto \exp(i\varphi \sin \theta)$ is continuous and 2π -periodic. Its Fourier coefficients (33) are absolutely summable by Lemma 9.3. A continuous function on the circle with absolutely summable Fourier coefficients equals its Fourier series pointwise on $(0, 2\pi]$. Both sides of (34) are 2π -periodic, so the identity extends to all real θ by reducing modulo 2π ($e^{in(\theta-2\pi m)} = e^{in\theta}$). $\square$

Corollary 9.5 (Discrete spectrum). *In the paper’s 2π Fourier convention, $e^{inkx} = e^{2\pi i q_n x}$ with $q_n = nk/(2\pi) = n/\Lambda$. The object is a discrete spectrum at those frequencies. Larger φ populates larger $|n|$, so pairs (q, q') leave the axes where $\Gamma_C \Gamma_S = 1$, and a separable CTF becomes inaccurate.*

10 Assembly

Theorem 10.1 (Working formula). *Under Approximation 2.10,*

$$R(\mathbf{q}, \mathbf{q}', \Delta z) = R_0(\mathbf{q}, \mathbf{q}', \Delta z, 0) E_S(\mathbf{q}, \mathbf{q}', \Delta z) E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}'), \quad (35)$$

with E_S from Theorem 5.3 and $E_{C,\text{tot}}$ from Theorem 6.5. Nac/ac specialisation is substitution of coefficients, not a new theorem.

Proof. Substitute (5) into (4) and use that the $(\mathbf{k}, \varepsilon)$ integral factors. $\square$

11 Correspondence with the Lean formalization

The identities above are encoded without `sorry` in this repository, using Mathlib’s Gaussian and complex-power lemmas. Wavelength is the field `lam` (Lean reserves λ).

Statement	Paper	Lean name
FWHM / $\sigma^2 = \Delta^2/(8 \ln 2)$	—	<code>gaussian_fwhm, variance_from_fwhm</code>
Char. fun. $E[e^{2\pi i a k}]$	A1	<code>charFun_source1D, charFun_source2D</code>
$\nabla \chi_S / \partial_q \chi_S$	—	<code>hasDerivAt_chiS, hasGradientAt_chiS2</code>
Taylor remainder $o(k)$	(3)	<code>chiS_taylor</code>
$b_1 \varepsilon + b_2 \varepsilon^2$	—	<code>chiC_sub</code>
$E_S = e^{-\kappa a^2}$	(5)	<code>spatialEnvelope_eq_fwhm</code>
$E_{C,\text{tot}}$ closed form	(6a)–(6b)	<code>chromaticEnvelopeIntegral_eq_closed</code>
Polar E_{CC}	A2	<code>ecc_eq_polar</code>
$\Gamma_S = e^{-2\kappa uv}$	(7b)	<code>gammaS_eq_kappa, gammaS_ac</code>
Axes / perfect coherence	—	<code>gammaS_axis, gamma_product_perfect</code>
$R_{\text{CTF}} = R_{\text{FO}} \Gamma_C \Gamma_S$	(7)	<code>R_CTF_eq_R_FO_mul_gamma</code>
Jacobi–Anger	§3.2	<code>jacobi_anger</code>

Modelling approximations 2.4 and 2.10 are *not* Lean theorems: the working kernel R_{FO} is defined as the factorised expression after those truncations.

12 Analytic pitfalls

1. **Principal square root.** Always $\exp(\frac{1}{2}\text{Log } z)$ with $\text{Arg} \in (-\pi, \pi]$. The identity $(A/z)^{1/2} = A^{1/2}z^{-1/2}$ requires $\text{Arg}(z) \neq \pi$. For $z = 1 - 4\pi i b_2 \sigma_E^2$ one has $\text{Re}(z) = 1$, so the cut is avoided and $b_2 = 0$ yields $+1$.
2. **Argument of $1 - iy$.** $\text{Arg}(1 - iy) = -\arctan y$ for *all* real y because the real part is identically 1. There is no branch jump as y crosses 0.
3. **Complex quadratic Gaussians.** [Lemma 3.4](#) requires $\text{Re}(B) < 0$, which comes only from the real Gaussian density. A purely imaginary quadratic coefficient would make the integral oscillatory.
4. **Sign of $-C^2/(4B)$.** $C = 2\pi i b_1$ implies $C^2 = -4\pi^2 b_1^2$. Forgetting $i^2 = -1$ produces a purely oscillatory “envelope”.
5. **Two-dimensional inner product.** $|\mathbf{a}|^2$ contains $\mathbf{q} \cdot \mathbf{q}'$, not qq' . Γ_S contains $\nabla \chi_S(\mathbf{q}) \cdot \nabla \chi_S(\mathbf{q}')$, not a product of Euclidean norms.
6. **Printed Γ_C has no extra conjugate.** On the axis $\mathbf{q} = \mathbf{0}$, only $|\Gamma_C| = 1$ in general for ac.
7. **FWHM versus variance.** $\kappa = \pi^2 q_{\text{ill}}^2 / (4 \ln 2)$ equals $2\pi^2 \sigma_{\text{ill}}^2$, not $\pi^2 \sigma^2$. Mixing FWHM and σ typically produces a factor-of-four error.
8. **Defocus sign.** $W = e^{+2\pi i \chi}$ with $+\frac{1}{2}\Delta z \lambda q^2$. Do not convert χ to radians and then insert another 2π .

References

- [1] K. M. Yu, K. L. W. Lau, M. S. Altman, *Fourier optics of image formation in aberration-corrected LEEM*, *Ultramicroscopy* **200** (2019) 160–168.
- [2] A. B. Pang, Th. Müller, M. S. Altman, E. Bauer, *J. Phys.: Condens. Matter* **21** (2009) 314006.
- [3] S. M. Schramm, A. B. Pang, M. S. Altman, R. M. Tromp, *Ultramicroscopy* **115** (2012) 88–108.
- [4] K. M. Yu, A. Locatelli, M. S. Altman, *Ultramicroscopy* **183** (2017) 109–116.
- [5] R. M. Tromp, S. M. Schramm, *Ultramicroscopy* (2012), optimization of the CTF.
- [6] K. J. Hanßen, L. Trepte, *Optik* **32** (1971) 519.